

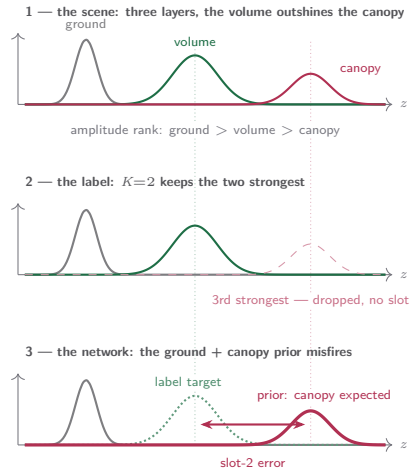
DLR-TomoSAR — Set prediction

the output is a set · Hungarian matching · the gated head · slot imbalance

July 2026

The small- K failure, concretely ($K=2$)

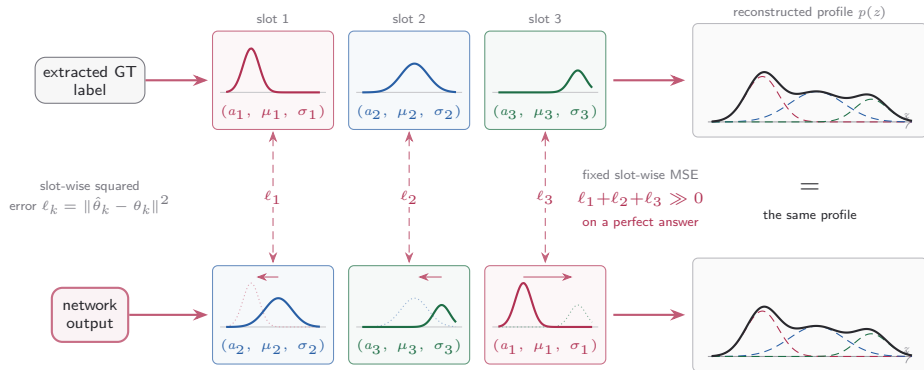
- Most pixels hold ground + canopy — the network internalises that prior: slot 2 *expects* a canopy.
- Here a mid-height volume outshines the canopy. GT extraction keeps the K strongest components, so the volume *becomes* the second Gaussian — the canopy leaves the label entirely.
- The prior misfires: slot 2 lands at canopy height while the label holds the volume — a **capacity failure of small K** : the label itself has no room for a third layer.
- At larger K a different failure appears: the sorting mechanism itself (two frames ahead).



the label has no room for a third layer — the miss is built into $K=2$

The output is a set, not a vector

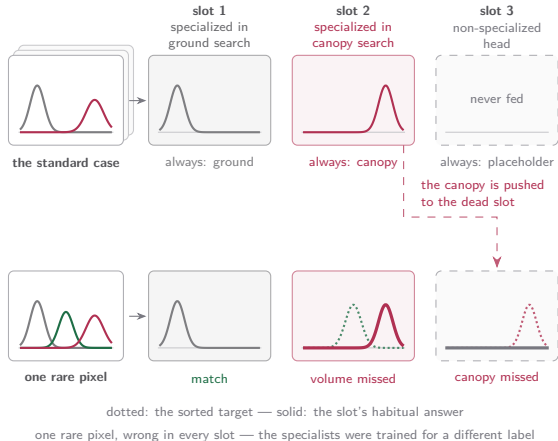
Per pixel the network fills K slots, each a triplet $\theta_k = (a_k, \mu_k, \sigma_k)$ — output channels with no intrinsic order.



This is the **permutation-invariance problem** — the loss must score the set, not the slot order.

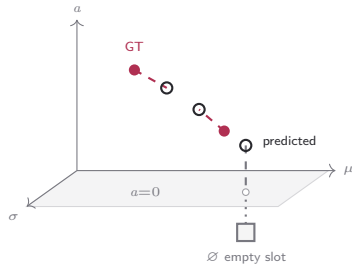
The sorting instability, concretely ($K=3$)

- Almost every pixel holds two layers. Under μ -sorting the head *specializes*: slot 1 learns the ground, slot 2 learns the canopy, and slot 3 almost never sees an active Gaussian — it learns silence.
- A three-layer pixel reshuffles the targets: the volume drops into slot 2 — the canopy specialist — and the canopy is pushed into slot 3 — the silent slot.
- Habits win: slot 2 emits its usual canopy and misses the volume; slot 3 stays silent and the canopy loses its predictor — the rare pixel is wrong in every slot at once.
- The failure is scoring *slots*; the way out is to score the *set* (next).



Hungarian matching

- Each Gaussian is a **point in (a, μ, σ) space**: prediction and GT are two point sets; the matcher pairs them by distance and scores the **set**, not the slot layout.



Filled = active GT, open = predicted, $\emptyset$ = empty GT slot (below $a=0$).
Its match (grey) is zero-weighted, so $\hat{\theta}_3$ is free.

$$\mathcal{L} = \min_{\pi \in S_K} \sum_{k=1}^K w_k \left\| \hat{\theta}_{\pi(k)} - \theta_k \right\|^2 \quad \theta_k = (a_k, \mu_k, \sigma_k)$$

$w_k = 1[a_k > \tau]$ — GT *active mask*: 1 if present, 0 if empty.

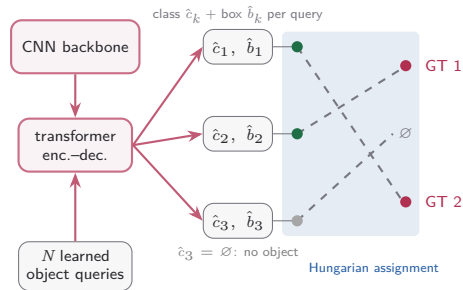
	GT θ_j			
	θ_1	θ_2	$\emptyset$	
$\hat{\theta}_1$	0.9	0.2	0.0	$\xrightarrow[\text{min cost}]{\pi^*}$
$\hat{\theta}_2$	0.4	0.8	0.0	
$\hat{\theta}_3$	0.6	0.5	0.0	
cost $C_{ij} = w_j \ \hat{\theta}_i - \theta_j\ ^2$				$\sum C_{\pi^*} = 0.6$

Column $\emptyset$ is a placeholder GT ($w_j=0$) — all-zero, so the leftover prediction matches it for free: $\hat{\theta}_1 \rightarrow \theta_2$, $\hat{\theta}_2 \rightarrow \theta_1$, $\hat{\theta}_3 \rightarrow \emptyset$.

Set-prediction lineage: DETR

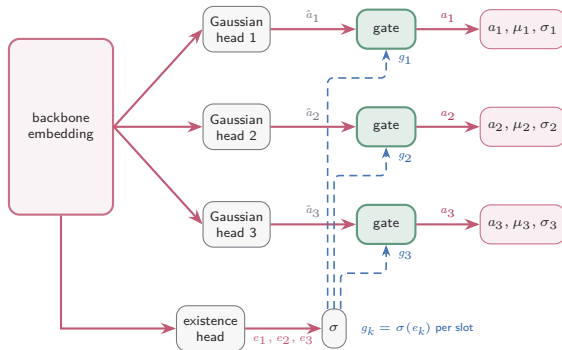
DETR (Carion et al., ECCV 2020) framed detection as **direct set prediction**: bipartite (Hungarian) matching + a per-element existence mechanism. **The answer to the small- K failure**: grow K until every layer fits, and let existence gates close the slots a pixel does not need.

- The gate replaces DETR's classification softmax: absence is encoded as *zero amplitude*, not a class — the $3K$ -channel contract is unchanged.
- Under sorted_gt the gate merely relabels slots — **pair with Hungarian**.



DETR: every query emits a class and a box; absence is the explicit class $\emptyset$ — unmatched queries are trained onto $\emptyset$ and leave the box loss.

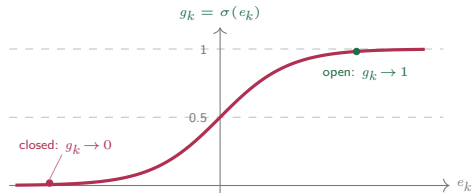
Inside the set-prediction head — gate, blend, match



each gate: $a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$
 μ_k, σ_k pass through $\cdot a_{\text{off}}$ learned per slot

The gate is a sigmoid on the existence logit e_k :

- Open slots ($g_k \rightarrow 1$): $a_k \approx \hat{a}_k$ — the raw regression survives.
- Closed slot ($g_k \rightarrow 0$): $a_k \rightarrow a_{\text{off},k} \approx 0$ whatever $\hat{a}_k$ is, collapsing onto the empty-set target at near-zero loss.
- e_k is internal — the head still emits $3K$ channels (a_k, μ_k, σ_k); a_{off} is a learned per-slot scalar (init 0).



Set prediction, controlled — the head leads at both K ; the matching matters only at $K=5$

metric	$K=2$				$K=5$			
	conv head		set-pred head		conv head		set-pred head	
	sort	Hung	sort	Hung	sort	Hung	sort	Hung
<i>reconstruction (curve)</i>								
curve R^2 $\uparrow$	0.569 $\pm$.008	0.603 $\pm$.012	0.536 $\pm$.026	0.578 $\pm$.016	0.561 $\pm$.019	0.587 $\pm$.012	0.534 $\pm$.007	0.566 $\pm$.016
curve MAE $\downarrow$	0.0496 $\pm$.0008	0.0471 $\pm$.0005	0.0503 $\pm$.0016	0.0480 $\pm$.0011	0.0554 $\pm$.0006	0.0521 $\pm$.0005	0.0565 $\pm$.0007	0.0537 $\pm$.0008
PSNR [dB] $\uparrow$	50.3 $\pm$.1	50.7 $\pm$.1	50.0 $\pm$.2	50.4 $\pm$.2	49.7 $\pm$.2	49.9 $\pm$.1	49.4 $\pm$.1	49.7 $\pm$.2
SSIM elev $\uparrow$	0.9586 $\pm$.0006	0.9601 $\pm$.0003	0.9586 $\pm$.0010	0.9598 $\pm$.0009	0.9530 $\pm$.0007	0.9565 $\pm$.0005	0.9527 $\pm$.0014	0.9553 $\pm$.0020
SSIM range $\uparrow$	0.9258 $\pm$.0021	0.9277 $\pm$.0019	0.9329 $\pm$.0039	0.9342 $\pm$.0028	0.9234 $\pm$.0028	0.9319 $\pm$.0039	0.9296 $\pm$.0020	0.9314 $\pm$.0026
SSIM azim $\uparrow$	0.9461 $\pm$.0012	0.9479 $\pm$.0007	0.9471 $\pm$.0020	0.9488 $\pm$.0014	0.9435 $\pm$.0011	0.9475 $\pm$.0011	0.9439 $\pm$.0013	0.9466 $\pm$.0016
profile cos med $\uparrow$	0.850 $\pm$.024	0.854 $\pm$.022	0.904 $\pm$.034	0.896 $\pm$.033	0.838 $\pm$.041	0.895 $\pm$.028	0.905 $\pm$.012	0.891 $\pm$.014
<i>detection (matched)</i>								
peak err mean $\downarrow$	4.70 $\pm$.47	4.80 $\pm$.52	1.66 $\pm$.09	1.63 $\pm$.08	4.83 $\pm$.71	2.58 $\pm$ 1.51	1.64 $\pm$.04	1.56 $\pm$.10
median $\downarrow$	0.81 $\pm$.30	1.07 $\pm$.37	0.67 $\pm$.00	0.67 $\pm$.00	0.81 $\pm$.30	0.67 $\pm$.00	0.67 $\pm$.00	0.67 $\pm$.00
p95 $\downarrow$	15.44 $\pm$.47	15.57 $\pm$.56	6.71 $\pm$.47	6.04 $\pm$.47	15.44 $\pm$.47	9.93 $\pm$ 4.26	7.38 $\pm$.00	5.77 $\pm$.37
precision $\uparrow$	0.821 $\pm$.008	0.822 $\pm$.002	0.816 $\pm$.005	0.813 $\pm$.004	0.620 $\pm$.011	0.636 $\pm$.038	0.627 $\pm$.009	0.573 $\pm$.027
recall $\uparrow$	0.641 $\pm$.029	0.642 $\pm$.045	0.887 $\pm$.008	0.893 $\pm$.004	0.648 $\pm$.040	0.802 $\pm$.093	0.867 $\pm$.006	0.886 $\pm$.003
matched F1 $\uparrow$	0.720 $\pm$.020	0.720 $\pm$.027	0.850 $\pm$.006	0.851 $\pm$.004	0.633 $\pm$.025	0.709 $\pm$.058	0.728 $\pm$.007	0.696 $\pm$.020
count exact $\uparrow$	0.595 $\pm$.034	0.590 $\pm$.044	0.852 $\pm$.004	0.841 $\pm$.004	0.430 $\pm$.050	0.574 $\pm$.131	0.651 $\pm$.004	0.588 $\pm$.062
under $\downarrow$	0.327 $\pm$.036	0.329 $\pm$.057	0.018 $\pm$.005	0.017 $\pm$.004	0.320 $\pm$.055	0.134 $\pm$.122	0.031 $\pm$.002	0.027 $\pm$.004
over $\downarrow$	0.078 $\pm$.004	0.081 $\pm$.012	0.130 $\pm$.005	0.143 $\pm$.003	0.251 $\pm$.008	0.292 $\pm$.021	0.318 $\pm$.006	0.385 $\pm$.065

runs: UNet · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 ; column groups = $K=2$ and $K=5$, within each head $\times$ matching = 2×2 ; five seeded replicas per arm, cells mean $\pm$ seed std; held-out test on the matching $K_{\max}$ labels (K groups are different label sets — compare within a group). **bold** = best in row within its K group, marked only where best and worst differ by more than 5%; **green** = the gate's detection lift; **red** = the over-count price and, at $K=5$, the sorted-GT break. Peak median/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

The head by scatterer count — the gate saturates $k=1$; recall fades up the $K=5$ ladder

metric	$K=2$				$K=5$			
	conv head		set-pred head		conv head		set-pred head	
	sort	Hung	sort	Hung	sort	Hung	sort	Hung
matched precision $\uparrow$	0.821 $\pm$.008	0.822 $\pm$.002	0.816 $\pm$.005	0.813 $\pm$.004	0.620 $\pm$.011	0.636 $\pm$.038	0.627 $\pm$.009	0.573 $\pm$.027
$k=1$	0.853 $\pm$.008	0.848 $\pm$.004	0.849 $\pm$.005	0.836 $\pm$.003	0.711 $\pm$.018	0.719 $\pm$.057	0.726 $\pm$.008	0.640 $\pm$.045
$k=2$	0.782 $\pm$.007	0.790 $\pm$.002	0.759 $\pm$.014	0.771 $\pm$.009	0.562 $\pm$.011	0.535 $\pm$.020	0.525 $\pm$.009	0.471 $\pm$.015
$k=3$	—	—	—	—	0.549 $\pm$.009	0.574 $\pm$.010	0.519 $\pm$.016	0.538 $\pm$.009
$k=4$	—	—	—	—	0.421 $\pm$.002	0.525 $\pm$.007	0.419 $\pm$.014	0.510 $\pm$.011
$k=5$	—	—	—	—	0.393 $\pm$.014	0.470 $\pm$.013	0.394 $\pm$.010	0.468 $\pm$.011
matched recall $\uparrow$	0.641 $\pm$.029	0.642 $\pm$.045	0.887 $\pm$.008	0.893 $\pm$.004	0.648 $\pm$.040	0.802 $\pm$.093	0.867 $\pm$.006	0.886 $\pm$.003
$k=1$	0.620 $\pm$.046	0.617 $\pm$.071	0.997 $\pm$.001	0.997 $\pm$.000	0.643 $\pm$.070	0.876 $\pm$.162	0.996 $\pm$.002	0.994 $\pm$.002
$k=2$	0.672 $\pm$.008	0.677 $\pm$.007	0.731 $\pm$.019	0.744 $\pm$.010	0.830 $\pm$.010	0.862 $\pm$.011	0.893 $\pm$.008	0.906 $\pm$.009
$k=3$	—	—	—	—	0.645 $\pm$.005	0.718 $\pm$.010	0.684 $\pm$.017	0.763 $\pm$.004
$k=4$	—	—	—	—	0.375 $\pm$.008	0.489 $\pm$.010	0.421 $\pm$.015	0.527 $\pm$.010
$k=5$	—	—	—	—	0.233 $\pm$.003	0.290 $\pm$.016	0.281 $\pm$.010	0.327 $\pm$.015
<i>count acc pred $k \uparrow$ — share of pixels claiming exactly k whose true count is k</i>								
pred $k=1$	0.948 $\pm$.007	0.951 $\pm$.013	0.975 $\pm$.006	0.977 $\pm$.004	0.948 $\pm$.010	0.965 $\pm$.011	0.973 $\pm$.005	0.978 $\pm$.008
pred $k=2$	0.733 $\pm$.010	0.725 $\pm$.026	0.651 $\pm$.007	0.631 $\pm$.005	0.315 $\pm$.016	0.262 $\pm$.039	0.251 $\pm$.006	0.173 $\pm$.061
pred $k=3$	—	—	—	—	0.124 $\pm$.008	0.100 $\pm$.018	0.095 $\pm$.008	0.050 $\pm$.011
pred $k=4$	—	—	—	—	0.072 $\pm$.006	0.051 $\pm$.011	0.047 $\pm$.005	0.024 $\pm$.006
pred $k=5$	—	—	—	—	0.102 $\pm$.012	0.081 $\pm$.004	0.075 $\pm$.005	0.063 $\pm$.009

runs: UNet · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 ; column groups = $K=2$ and $K=5$ (head $\times$ matching within each); five seeded replicas per arm, cells mean $\pm$ seed std; held-out test; k = true scatterer count of the pixel; — = rank absent from the $K=2$ label set. **bold** = best in row within its K group, marked only where best and worst differ by more than 5%; **green** = the gate's $k=1$ saturation and the higher-rung wins, **red** = the layover price and the unreliable claimed pairs. Precision rewards under-prediction — rank on recall.

The head components — layover pays at $K=2$, the crowd pays at $K=5$

metric	$K=2$				$K=5$			
	conv head		set-pred head		conv head		set-pred head	
	sort	Hung	sort	Hung	sort	Hung	sort	Hung
matched μ MAE $\downarrow$	2.30 $\pm$.08	2.29 $\pm$.10	2.18 $\pm$.09	2.19 $\pm$.08	2.30 $\pm$.14	1.90 $\pm$.12	2.05 $\pm$.06	1.94 $\pm$.06
$k=1$	0.59 $\pm$.04	0.56 $\pm$.05	0.59 $\pm$.10	0.62 $\pm$.11	0.69 $\pm$.09	0.63 $\pm$.04	0.65 $\pm$.05	0.75 $\pm$.08
$k=2$	4.05 $\pm$.08	4.06 $\pm$.02	4.54 $\pm$.11	4.51 $\pm$.13	1.81 $\pm$.04	1.83 $\pm$.09	1.99 $\pm$.05	1.83 $\pm$.09
$k=3$	—	—	—	—	4.24 $\pm$.10	3.63 $\pm$.16	4.32 $\pm$.11	3.60 $\pm$.14
$k=4$	—	—	—	—	7.88 $\pm$.20	6.48 $\pm$.13	7.50 $\pm$.23	6.55 $\pm$.25
$k=5$	—	—	—	—	9.56 $\pm$.28	9.16 $\pm$.23	9.43 $\pm$.18	9.48 $\pm$.26
matched σ MAE $\downarrow$	0.87 $\pm$.03	0.87 $\pm$.04	0.82 $\pm$.03	0.83 $\pm$.03	0.79 $\pm$.04	0.79 $\pm$.05	0.70 $\pm$.01	0.77 $\pm$.03
$k=1$	0.25 $\pm$.01	0.26 $\pm$.02	0.23 $\pm$.01	0.24 $\pm$.01	0.35 $\pm$.03	0.36 $\pm$.04	0.29 $\pm$.01	0.35 $\pm$.03
$k=2$	1.51 $\pm$.01	1.50 $\pm$.01	1.70 $\pm$.06	1.71 $\pm$.06	0.97 $\pm$.01	1.11 $\pm$.04	1.00 $\pm$.00	1.12 $\pm$.04
$k=3$	—	—	—	—	1.36 $\pm$.01	1.45 $\pm$.05	1.41 $\pm$.01	1.46 $\pm$.03
$k=4$	—	—	—	—	1.60 $\pm$.01	1.70 $\pm$.08	1.70 $\pm$.00	1.70 $\pm$.04
$k=5$	—	—	—	—	1.66 $\pm$.01	1.84 $\pm$.13	1.83 $\pm$.01	1.81 $\pm$.07
matched a MAE $\downarrow$	0.374 $\pm$.026	0.380 $\pm$.033	0.306 $\pm$.002	0.311 $\pm$.004	0.526 $\pm$.026	0.564 $\pm$.050	0.442 $\pm$.008	0.552 $\pm$.044
$k=1$	0.161 $\pm$.017	0.180 $\pm$.026	0.114 $\pm$.004	0.133 $\pm$.005	0.172 $\pm$.015	0.230 $\pm$.040	0.130 $\pm$.007	0.235 $\pm$.058
$k=2$	0.593 $\pm$.022	0.585 $\pm$.023	0.590 $\pm$.006	0.573 $\pm$.011	0.725 $\pm$.023	0.906 $\pm$.053	0.744 $\pm$.014	0.920 $\pm$.048
$k=3$	—	—	—	—	1.018 $\pm$.020	1.091 $\pm$.016	1.024 $\pm$.013	1.096 $\pm$.030
$k=4$	—	—	—	—	1.067 $\pm$.015	1.078 $\pm$.026	1.038 $\pm$.011	1.049 $\pm$.026
$k=5$	—	—	—	—	0.988 $\pm$.015	1.001 $\pm$.041	0.895 $\pm$.007	0.915 $\pm$.041

runs: UNet · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 ; column groups = $K=2$ and $K=5$ (head $\times$ matching within each); five seeded replicas per arm, cells mean $\pm$ seed std; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only; — = rank absent from the $K=2$ label set. **bold** = best in row within its K group, marked only where best and worst differ by more than 5%; **green** = the gate's amplitude win at $K=2$ and the Hungarian μ buy at $K=5$; **red** = the layover price at $K=2$ and the amplitude Hungarian pays at $K=5$. μ , σ in elevation units; a in linear amplitude.

The $K=2$ head, predicted statistics vs the GT — the gate calibrates activity, pays in amplitude

		conv head		set-pred head	
prediction mean	GT	sort	Hung	sort	Hung
slot 0 — strongest scatterer; GT-active on every pixel					
active frac	1.00	0.69 ±.03	0.68 ±.05	0.990 ±.003	0.990 ±.002
a — active	0.77	0.90 ±.03	0.87 ±.05	0.61 ±.01	0.58 ±.01
μ — active	−4.89	−3.92 ±.10	−3.66 ±.13	−4.38 ±.22	−4.13 ±.14
σ — active	1.76	1.48 ±.05	1.50 ±.07	1.34 ±.03	1.38 ±.03
slot 1 — second scatterer; GT-active on 26% of pixels					
active frac	0.260	0.299 ±.006	0.304 ±.015	0.380 ±.008	0.394 ±.005
a — active	1.43	1.04 ±.03	1.10 ±.02	0.89 ±.02	0.94 ±.03
a — inact.	0.00	0.42 ±.01	0.52 ±.07	0.30 ±.02	0.39 ±.04
μ — active	11.34	10.09 ±.33	9.15 ±.26	9.77 ±.50	8.80 ±.32
μ — inact.	—	4.09 ±.31	2.04 ±.75	2.57 ±.37	−0.29 ±.37
σ — active	3.53	2.94 ±.05	2.98 ±.06	2.80 ±.08	2.78 ±.05
σ — inact.	—	1.87 ±.03	1.76 ±.05	1.76 ±.03	1.70 ±.03
distribution width — std over GT-active pixels					
a — slot 0	1.48	1.17 ±.03	1.16 ±.03	0.91 ±.01	0.93 ±.03
μ — slot 0	3.30	1.99 ±.07	2.13 ±.11	2.00 ±.08	2.23 ±.16
σ — slot 0	2.84	0.95 ±.03	1.03 ±.04	0.93 ±.03	1.07 ±.03
a — slot 1	1.45	0.83 ±.04	0.91 ±.02	0.77 ±.03	0.82 ±.03
μ — slot 1	16.24	8.72 ±.18	8.88 ±.13	8.86 ±.74	9.47 ±.21
σ — slot 1	4.01	1.07 ±.02	1.24 ±.05	1.04 ±.05	1.15 ±.05

runs: UNet · $K=2$ · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 (head $\times$ matching = column); five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = the gate's activity and leakage wins, **red** = the slot-1 over-fire and the amplitude payment. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; a in linear amplitude.

The $K=5$ head, predicted statistics vs the GT — slot 0 calibrated, the tail over-fired

		conv head		set-pred head	
prediction mean	GT	sort	Hung	sort	Hung
predicted active fraction — share of pixels where the slot fires					
slot 0	1.000	0.701 $\pm$.054	0.880 $\pm$.126	0.989 $\pm$.002	0.983 $\pm$.004
slot 1	0.239	0.319 $\pm$.009	0.278 $\pm$.050	0.385 $\pm$.005	0.324 $\pm$.042
slot 2	0.085	0.218 $\pm$.006	0.193 $\pm$.065	0.263 $\pm$.003	0.234 $\pm$.046
slot 3	0.035	0.137 $\pm$.005	0.169 $\pm$.050	0.162 $\pm$.013	0.239 $\pm$.056
slot 4	0.015	0.060 $\pm$.015	0.207 $\pm$.068	0.099 $\pm$.009	0.349 $\pm$.087
a — predicted mean over GT-active pixels					
slot 0	0.84	0.99 $\pm$.08	0.64 $\pm$.14	0.67 $\pm$.01	0.51 $\pm$.06
slot 1	1.67	0.94 $\pm$.04	0.67 $\pm$.07	0.80 $\pm$.02	0.67 $\pm$.05
slot 2	1.41	0.30 $\pm$.02	0.71 $\pm$.07	0.26 $\pm$.02	0.57 $\pm$.05
slot 3	0.97	0.14 $\pm$.01	0.28 $\pm$.05	0.12 $\pm$.01	0.23 $\pm$.02
slot 4	0.66	0.09 $\pm$.01	0.21 $\pm$.01	0.07 $\pm$.01	0.19 $\pm$.03
a — predicted mean over GT-inactive pixels (leakage)					
slot 1	0.00	0.65 $\pm$.02	0.26 $\pm$.21	0.46 $\pm$.01	0.39 $\pm$.21
slot 2	0.00	0.37 $\pm$.01	0.40 $\pm$.18	0.29 $\pm$.00	0.40 $\pm$.12
slot 3	0.00	0.17 $\pm$.01	0.56 $\pm$.25	0.14 $\pm$.00	0.31 $\pm$.20
slot 4	0.00	0.11 $\pm$.01	0.58 $\pm$.39	0.07 $\pm$.01	0.41 $\pm$.31
distribution width — std over GT-active pixels					
a — slot 0	1.59	1.33 $\pm$.07	1.09 $\pm$.09	1.04 $\pm$.02	0.92 $\pm$.04
μ — slot 0	3.02	2.47 $\pm$.06	2.37 $\pm$.05	2.28 $\pm$.04	2.39 $\pm$.12
σ — slot 0	1.72	0.80 $\pm$.02	0.71 $\pm$.11	0.79 $\pm$.04	0.86 $\pm$.10
a — slot 1	1.60	0.82 $\pm$.02	0.87 $\pm$.05	0.75 $\pm$.01	0.83 $\pm$.02
μ — slot 1	11.09	8.44 $\pm$.10	8.83 $\pm$.10	8.48 $\pm$.10	8.98 $\pm$.15
σ — slot 1	2.30	0.87 $\pm$.02	1.28 $\pm$.03	0.88 $\pm$.01	1.29 $\pm$.04

runs: UNet · $K=5$ · param-MSE · presence none · full norm ladder · hv flips · patch 64×32 (head $\times$ matching = 2×2); five seeded replicas per arm, cells mean $\pm$ seed std; held-out test; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = the calibrated slot-0 gate, **red** = the over-fired tail slots. Width rows: std of the prediction over GT-active pixels vs the GT's own std, shown for slots 0-1 (tail slots in the report). μ , σ in elevation units; a in linear amplitude.

The imbalance failure (high K)

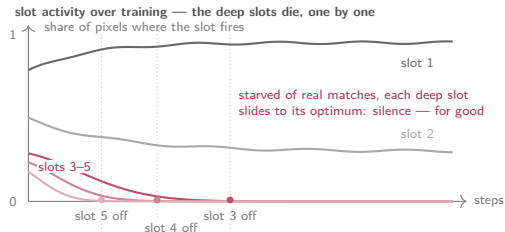
- Hungarian + existence gates make large K trainable — and expose the third failure: **most slots have nothing to match**.
- The matcher hands slot k a real scatterer (amplitude a^+) in a fraction ρ_k of cells, silence in the rest — deep slots see $\rho_k \approx 0.01$. Mean-reduced, the slot's amplitude minimizes the two-point mixture on the right (param-MSE).
- The push outweighs the pull 99:1; descent parks the slot at 1% of the real amplitude — **shrinking to effective silence is the loss optimum**, not a training accident.
- The working $K=2$ configuration mostly sidesteps the tail; the $K=5$ development confronts it head-on — the answer (next).

$$\ell_k(\hat{a}) = (\hat{a} - a_k^{\text{match}})^2, \quad a_k^{\text{match}} = \begin{cases} a^+ & \text{with prob. } \rho_k \\ 0 & \text{with prob. } 1 - \rho_k \end{cases}$$

$$\mathbb{E}[\ell_k(\hat{a})] = \rho_k \mathbb{E}[(\hat{a} - a^+)^2] + (1 - \rho_k) \hat{a}^2$$

$$\partial_{\hat{a}} \mathbb{E}[\ell_k] = \underbrace{2\rho_k (\hat{a} - \mathbb{E}[a^+])}_{\text{pull toward the target}} + \underbrace{2(1 - \rho_k) \hat{a}}_{\text{push toward silence}}$$

$$= 2(\hat{a} - \rho_k \mathbb{E}[a^+]) \implies \hat{a}_k^* = \rho_k \mathbb{E}[a^+] \approx 0.01 \mathbb{E}[a^+]$$



The answer to slot imbalance — lever 1: active normalization

Active slot = the GT placed a real scatterer in it:

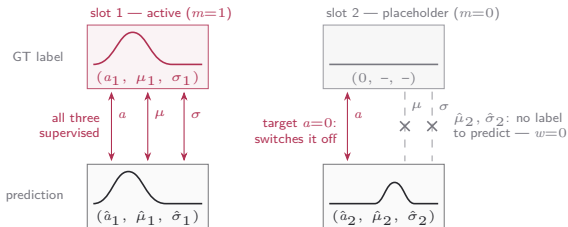
$$m_{p,k} = \mathbf{1}[a_{p,k}^{\text{GT}} > \tau]$$

pixel p , slot k ; $\tau = 10^{-3}$ linear amplitude — empty slots carry $a=0$. A *pixel* is active when any of its K slots is.

The lever — one reduction, dividing by the **weight mass**, not the cell count:

$$\ell = \frac{\sum_{p,k} w_{p,k} e_{p,k}}{\sum_{p,k} w_{p,k}}$$

e = matched error; w per term: a everywhere, μ, σ masked by m — nothing to tune.



$$\ell_a = \frac{\sum_{p,k} (\hat{a} - a^{\text{GT}})^2}{N_{\text{pix}} K}$$

$$\ell_\mu = \frac{\sum_{p,k} m (\hat{\mu} - \mu^{\text{GT}})^2}{N_{\text{pix}} K} \rightarrow \frac{\sum_{p,k} m (\hat{\mu} - \mu^{\text{GT}})^2}{\sum_{p,k} m}$$

$$\ell_\sigma = \frac{\sum_{p,k} m (\hat{\sigma} - \sigma^{\text{GT}})^2}{N_{\text{pix}} K} \rightarrow \frac{\sum_{p,k} m (\hat{\sigma} - \sigma^{\text{GT}})^2}{\sum_{p,k} m}$$

The answer to slot imbalance — lever 2: presence balance

$$\begin{aligned} \mathbb{E}[w \ell_k] &= f_k \frac{0.5}{f_k} \mathbb{E}[(\hat{a} - a^+)^2] + (1 - f_k) \frac{0.5}{1 - f_k} \hat{a}^2 \\ &= \frac{1}{2} \mathbb{E}[(\hat{a} - a^+)^2] + \frac{1}{2} \hat{a}^2 \end{aligned}$$

$$\partial_{\hat{a}} \mathbb{E}[w \ell_k] = \hat{a} - \mathbb{E}[a^+] + \hat{a}$$

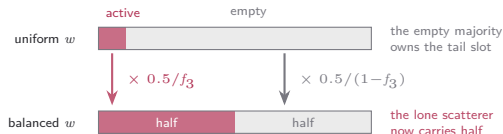
$$\hat{a}_k^* : \rho_k \mathbb{E}[a^+] \rightarrow \frac{1}{2} \mathbb{E}[a^+]$$

- **The imbalance is per slot:** sorting stacks the scatterers early — slot 1 nearly always active, the tail nearly never; one batch-wide fraction cannot see the tail.
- **Presence balance — inverse class frequency, slot by slot:** within every slot actives and empties carry half each:

$$w_{p,k} = \frac{0.5}{f_k} m_{p,k} + \frac{0.5}{1 - f_k} (1 - m_{p,k})$$

$$f_k = \text{clip}(\langle m_{p,k} \rangle_p, \varepsilon, 1 - \varepsilon)$$

m as on lever 1; f_k measured per batch ($\approx \rho_k$), clipped at $\varepsilon = 10^{-3}$.



The two levers — A stays free; B costs recall, then the curve

metric	K=2				K=5			
	none	+A	+B	+AB	none	+A	+B	+AB
<i>reconstruction (curve)</i>								
curve R^2 ↑	0.578 ±.016	0.579 ±.017	0.553 ±.011	0.558 ±.014	0.566 ±.016	0.572 ±.010	0.473 ±.040	0.481 ±.029
curve MAE ↓	0.0480 ±.0011	0.0479 ±.0008	0.0493 ±.0008	0.0489 ±.0007	0.0537 ±.0008	0.0533 ±.0007	0.0646 ±.0028	0.0637 ±.0025
PSNR [dB] ↑	50.41 ±.17	50.43 ±.17	50.16 ±.11	50.21 ±.14	49.71 ±.16	49.77 ±.11	48.88 ±.33	48.94 ±.25
SSIM elev ↑	0.9598 ±.0009	0.9600 ±.0007	0.9589 ±.0007	0.9592 ±.0006	0.9553 ±.0020	0.9559 ±.0014	0.9388 ±.0032	0.9398 ±.0025
SSIM range ↑	0.9342 ±.0028	0.9346 ±.0025	0.9308 ±.0027	0.9319 ±.0018	0.9314 ±.0026	0.9326 ±.0021	0.9084 ±.0050	0.9103 ±.0035
SSIM azim ↑	0.9488 ±.0014	0.9491 ±.0011	0.9466 ±.0011	0.9471 ±.0008	0.9466 ±.0016	0.9473 ±.0012	0.9288 ±.0036	0.9300 ±.0033
profile cos med ↑	0.896 ±.033	0.892 ±.042	0.883 ±.038	0.891 ±.035	0.891 ±.014	0.899 ±.030	0.859 ±.027	0.865 ±.035
peak err mean ↓	1.63 ±.08	1.63 ±.11	1.72 ±.16	1.67 ±.10	1.56 ±.10	1.50 ±.10	1.89 ±.12	1.85 ±.16
median ↓	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00	0.67 ±.00
p95 ↓	6.04 ±.47	6.31 ±.37	6.44 ±.60	6.31 ±.37	5.77 ±.37	5.77 ±.37	7.92 ±.88	7.79 ±.60
<i>detection (matched)</i>								
precision ↑	0.813 ±.004	0.816 ±.003	0.813 ±.011	0.814 ±.009	0.573 ±.027	0.575 ±.019	0.483 ±.036	0.502 ±.027
recall ↑	0.893 ±.004	0.893 ±.003	0.879 ±.007	0.882 ±.004	0.886 ±.003	0.889 ±.004	0.878 ±.011	0.880 ±.003
matched F1 ↑	0.851 ±.004	0.853 ±.002	0.845 ±.008	0.846 ±.005	0.696 ±.020	0.698 ±.013	0.622 ±.029	0.639 ±.022
count exact ↑	0.841 ±.004	0.841 ±.009	0.843 ±.022	0.844 ±.017	0.588 ±.062	0.589 ±.063	0.445 ±.202	0.529 ±.122
under ↓	0.017 ±.004	0.019 ±.004	0.026 ±.008	0.023 ±.002	0.027 ±.004	0.025 ±.005	0.025 ±.010	0.023 ±.004
over ↓	0.142 ±.003	0.140 ±.006	0.131 ±.017	0.133 ±.015	0.385 ±.065	0.386 ±.067	0.529 ±.209	0.448 ±.126
<i>seed-to-seed spread (std over 5 seeds) ↓</i>								
curve R^2	.016	.017	.011	.014	.016	.010	.040	.029
curve MAE	.0011	.0008	.0008	.0007	—	—	—	—
SSIM range	.0028	.0025	.0027	.0018	—	—	—	—
profile cos med	.033	.042	.038	.035	—	—	—	—
count exact	—	—	—	—	.062	.063	.202	.122
count over	—	—	—	—	.065	.067	.209	.126

runs: UNet · set-pred · Hungarian · param-MSE · full norm ladder · hv flips · patch 64×32; column groups = K=2 and K=5 (presence lever within each); five seeded replicas per arm, cells mean ± seed std; held-out test on the matching $K_{\max}$ labels. +A active normalization, +B presence balance, +AB both; each none arm = its K 's set-pred-Hungarian heads arm. **bold** = best in row within its K group, marked only where best and worst differ by more than 5%; spread rows compare the seed stds themselves (— = spread not reported for that K); **red** = B's recall price at K=2, its curve and precision price and seed destabilization at K=5 — A stays free at both. Peak median/p95: per-pixel peak-error distribution; profile cos med: median per-pixel cosine to the GT profile.

The levers by scatterer count — B's cost starts at the second scatterer and spreads at $K=5$

metric	$K=2$				$K=5$			
	none	+A	+B	+AB	none	+A	+B	+AB
matched precision $\uparrow$	0.813 $\pm$.004	0.816 $\pm$.003	0.813 $\pm$.011	0.814 $\pm$.009	0.573 $\pm$.027	0.575 $\pm$.019	0.483 $\pm$.036	0.502 $\pm$.027
$k=1$	0.836 $\pm$.003	0.839 $\pm$.005	0.847 $\pm$.017	0.846 $\pm$.015	0.640 $\pm$.045	0.643 $\pm$.036	0.521 $\pm$.067	0.554 $\pm$.049
$k=2$	0.771 $\pm$.009	0.776 $\pm$.009	0.754 $\pm$.007	0.757 $\pm$.008	0.471 $\pm$.015	0.473 $\pm$.008	0.415 $\pm$.007	0.416 $\pm$.012
$k=3$	—	—	—	—	0.538 $\pm$.009	0.540 $\pm$.007	0.478 $\pm$.005	0.478 $\pm$.012
$k=4$	—	—	—	—	0.510 $\pm$.011	0.510 $\pm$.007	0.466 $\pm$.013	0.469 $\pm$.010
$k=5$	—	—	—	—	0.468 $\pm$.011	0.467 $\pm$.005	0.439 $\pm$.017	0.441 $\pm$.006
matched recall $\uparrow$	0.893 $\pm$.004	0.893 $\pm$.003	0.879 $\pm$.007	0.882 $\pm$.004	0.886 $\pm$.003	0.889 $\pm$.004	0.878 $\pm$.011	0.880 $\pm$.003
$k=1$	0.997 $\pm$.000	0.997 $\pm$.001	0.992 $\pm$.009	0.996 $\pm$.001	0.994 $\pm$.002	0.996 $\pm$.002	0.987 $\pm$.008	0.991 $\pm$.003
$k=2$	0.744 $\pm$.010	0.744 $\pm$.007	0.719 $\pm$.007	0.721 $\pm$.010	0.906 $\pm$.009	0.908 $\pm$.011	0.902 $\pm$.016	0.897 $\pm$.007
$k=3$	—	—	—	—	0.763 $\pm$.004	0.767 $\pm$.005	0.734 $\pm$.016	0.733 $\pm$.012
$k=4$	—	—	—	—	0.527 $\pm$.010	0.535 $\pm$.011	0.514 $\pm$.027	0.519 $\pm$.014
$k=5$	—	—	—	—	0.327 $\pm$.015	0.339 $\pm$.013	0.336 $\pm$.033	0.344 $\pm$.016
count acc pred $k \uparrow$ — share of pixels claiming exactly k whose true count is k								
pred $k=1$	0.977 $\pm$.004	0.975 $\pm$.006	0.970 $\pm$.005	0.970 $\pm$.005	0.978 $\pm$.008	0.977 $\pm$.006	0.972 $\pm$.012	0.976 $\pm$.005
pred $k=2$	0.631 $\pm$.005	0.634 $\pm$.012	0.646 $\pm$.030	0.643 $\pm$.027	0.173 $\pm$.061	0.161 $\pm$.050	0.108 $\pm$.070	0.116 $\pm$.051
pred $k=3$	—	—	—	—	0.050 $\pm$.011	0.057 $\pm$.009	0.032 $\pm$.008	0.033 $\pm$.005
pred $k=4$	—	—	—	—	0.024 $\pm$.006	0.023 $\pm$.004	0.009 $\pm$.003	0.010 $\pm$.002
pred $k=5$	—	—	—	—	0.063 $\pm$.009	0.065 $\pm$.006	0.039 $\pm$.003	0.041 $\pm$.005

runs: UNet · set-pred · Hungarian · param-MSE · full norm ladder · hv flips · patch 64×32 ; column groups = $K=2$ and $K=5$ (presence lever within each); five seeded replicas per arm, cells mean $\pm$ seed std; held-out test; k = true scatterer count of the pixel; — = rank absent from the $K=2$ label set. +A active normalization, +B presence balance, +AB both. **bold** = best in row within its K group, marked only where best and worst differ by more than 5%; **red** = B's $k=2$ cost at $K=2$, its precision price and the wrong-way $k=3$ at $K=5$. Precision rewards under-prediction — rank on recall.

The lever components — B is noise at $K=2$; at $K=5$ it buys amplitude, pays position and width

metric	$K=2$				$K=5$			
	none	+A	+B	+AB	none	+A	+B	+AB
matched μ MAE ↓	2.19 ±.08	2.15 ±.09	2.27 ±.12	2.23 ±.07	1.94 ±.06	1.97 ±.07	2.14 ±.07	2.11 ±.12
$k=1$	0.62 ±.11	0.63 ±.15	0.66 ±.15	0.63 ±.15	0.75 ±.08	0.71 ±.14	0.88 ±.10	0.82 ±.12
$k=2$	4.51 ±.13	4.41 ±.19	4.64 ±.16	4.60 ±.11	1.83 ±.09	1.90 ±.05	1.99 ±.08	1.99 ±.11
$k=3$	—	—	—	—	3.60 ±.14	3.72 ±.07	3.86 ±.14	3.87 ±.15
$k=4$	—	—	—	—	6.55 ±.25	6.59 ±.25	6.67 ±.17	6.63 ±.18
$k=5$	—	—	—	—	9.48 ±.26	9.40 ±.23	9.71 ±.17	9.64 ±.10
matched σ MAE ↓	0.83 ±.03	0.82 ±.04	0.83 ±.02	0.81 ±.02	0.77 ±.03	0.78 ±.02	0.83 ±.02	0.85 ±.04
$k=1$	0.24 ±.01	0.24 ±.01	0.24 ±.01	0.23 ±.01	0.35 ±.03	0.35 ±.03	0.39 ±.02	0.41 ±.04
$k=2$	1.71 ±.06	1.69 ±.09	1.70 ±.06	1.67 ±.04	1.12 ±.04	1.13 ±.03	1.18 ±.03	1.20 ±.05
$k=3$	—	—	—	—	1.46 ±.03	1.47 ±.03	1.54 ±.02	1.56 ±.04
$k=4$	—	—	—	—	1.70 ±.04	1.73 ±.03	1.76 ±.06	1.79 ±.03
$k=5$	—	—	—	—	1.81 ±.07	1.87 ±.05	1.83 ±.08	1.87 ±.04
matched a MAE ↓	0.311 ±.004	0.315 ±.009	0.317 ±.012	0.316 ±.012	0.552 ±.044	0.540 ±.038	0.485 ±.019	0.469 ±.014
$k=1$	0.133 ±.005	0.134 ±.010	0.136 ±.012	0.133 ±.009	0.235 ±.058	0.228 ±.053	0.231 ±.020	0.219 ±.008
$k=2$	0.573 ±.011	0.582 ±.008	0.585 ±.012	0.589 ±.019	0.920 ±.048	0.901 ±.038	0.822 ±.048	0.790 ±.030
$k=3$	—	—	—	—	1.096 ±.030	1.075 ±.020	0.901 ±.028	0.892 ±.039
$k=4$	—	—	—	—	1.049 ±.026	1.029 ±.020	0.800 ±.041	0.798 ±.049
$k=5$	—	—	—	—	0.915 ±.041	0.883 ±.033	0.638 ±.050	0.627 ±.050

runs: UNet · set-pred · Hungarian · param-MSE · full norm ladder · hv flips · patch 64×32 ; column groups = $K=2$ and $K=5$ (presence lever within each); five seeded replicas per arm, cells mean ± seed std; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only; — = rank absent from the $K=2$ label set. +A active normalization, +B presence balance, +AB both. **bold** = best in row within its K group, marked only where best and worst differ by more than 5%; **green** = the amplitude B buys at $K=5$; **red** = the position and width it pays. μ , σ in elevation units; a in linear amplitude.

The $K=2$ levers, predicted statistics vs the GT — the head sets the distribution, B trims the empties

prediction mean	GT	none	+A	+B	+AB
<i>slot 0 — strongest scatterer; GT-active on every pixel</i>					
active frac	1.00	0.990 $\pm$.002	0.989 $\pm$.004	0.983 $\pm$.008	0.986 $\pm$.004
a — active	0.77	0.58 $\pm$.01	0.58 $\pm$.02	0.58 $\pm$.01	0.58 $\pm$.01
μ — active	-4.89	-4.13 $\pm$.14	-4.15 $\pm$.18	-3.91 $\pm$.18	-3.93 $\pm$.16
σ — active	1.76	1.38 $\pm$.03	1.37 $\pm$.02	1.34 $\pm$.03	1.35 $\pm$.02
<i>slot 1 — second scatterer; GT-active on 26% of pixels</i>					
active frac	0.260	0.394 $\pm$.005	0.389 $\pm$.004	0.380 $\pm$.016	0.381 $\pm$.013
a — active	1.43	0.94 $\pm$.03	0.93 $\pm$.03	0.96 $\pm$.01	0.94 $\pm$.03
a — inact.	0.00	0.39 $\pm$.04	0.40 $\pm$.04	0.35 $\pm$.04	0.34 $\pm$.02
μ — active	11.34	8.80 $\pm$.32	8.65 $\pm$.40	8.89 $\pm$.48	8.63 $\pm$.44
μ — inact.	—	-0.29 $\pm$.37	-0.37 $\pm$.40	1.23 $\pm$.68	1.09 $\pm$.56
σ — active	3.53	2.78 $\pm$.05	2.77 $\pm$.06	2.73 $\pm$.10	2.69 $\pm$.09
σ — inact.	—	1.70 $\pm$.03	1.70 $\pm$.05	1.62 $\pm$.07	1.60 $\pm$.08
<i>distribution width — std over GT-active pixels</i>					
a — slot 0	1.48	0.93 $\pm$.03	0.92 $\pm$.03	0.92 $\pm$.03	0.93 $\pm$.03
μ — slot 0	3.30	2.23 $\pm$.16	2.21 $\pm$.17	2.49 $\pm$.07	2.44 $\pm$.16
σ — slot 0	2.84	1.07 $\pm$.03	1.05 $\pm$.03	1.01 $\pm$.04	1.01 $\pm$.04
a — slot 1	1.45	0.82 $\pm$.03	0.81 $\pm$.02	0.84 $\pm$.02	0.83 $\pm$.02
μ — slot 1	16.24	9.47 $\pm$.21	9.41 $\pm$.62	9.00 $\pm$.55	8.88 $\pm$.45
σ — slot 1	4.01	1.15 $\pm$.05	1.15 $\pm$.07	1.14 $\pm$.06	1.12 $\pm$.07

runs: UNet · set-pred · Hungarian · $K=2$ · param-MSE · full norm ladder · hv flips · patch 64×32 (presence lever = column); five seeded replicas per arm, cells mean $\pm$ seed std; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = the empty-slot trim B buys. Inactive GT slots define no μ/σ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. μ , σ in elevation units; a in linear amplitude.

The $K=5$ levers, predicted statistics vs the GT — B re-brightens the tail, then over-fires it

prediction mean	GT	none	+A	+B	+AB
<i>predicted active fraction — share of pixels where the slot fires</i>					
slot 0	1.000	0.983 $\pm$.004	0.987 $\pm$.003	0.983 $\pm$.008	0.986 $\pm$.003
slot 1	0.239	0.324 $\pm$.042	0.325 $\pm$.037	0.349 $\pm$.030	0.347 $\pm$.022
slot 2	0.085	0.234 $\pm$.046	0.233 $\pm$.054	0.335 $\pm$.032	0.327 $\pm$.025
slot 3	0.035	0.239 $\pm$.056	0.235 $\pm$.047	0.432 $\pm$.229	0.339 $\pm$.052
slot 4	0.015	0.349 $\pm$.087	0.345 $\pm$.081	0.410 $\pm$.097	0.414 $\pm$.131
<i>a — predicted mean over GT-active pixels</i>					
slot 0	0.84	0.51 $\pm$.06	0.52 $\pm$.06	0.53 $\pm$.03	0.55 $\pm$.01
slot 1	1.67	0.67 $\pm$.05	0.67 $\pm$.07	0.85 $\pm$.04	0.87 $\pm$.03
slot 2	1.41	0.57 $\pm$.05	0.58 $\pm$.05	0.76 $\pm$.04	0.71 $\pm$.06
slot 3	0.97	0.23 $\pm$.02	0.23 $\pm$.03	0.59 $\pm$.06	0.58 $\pm$.06
slot 4	0.66	0.19 $\pm$.03	0.18 $\pm$.03	0.54 $\pm$.05	0.53 $\pm$.05
<i>a — predicted mean over GT-inactive pixels (leakage)</i>					
slot 1	0.00	0.39 $\pm$.21	0.37 $\pm$.18	0.26 $\pm$.07	0.25 $\pm$.08
slot 2	0.00	0.40 $\pm$.12	0.40 $\pm$.11	0.40 $\pm$.02	0.38 $\pm$.07
slot 3	0.00	0.31 $\pm$.20	0.32 $\pm$.22	0.43 $\pm$.10	0.45 $\pm$.06
slot 4	0.00	0.41 $\pm$.31	0.39 $\pm$.30	0.51 $\pm$.10	0.54 $\pm$.10
<i>distribution width — std over GT-active pixels</i>					
a — slot 0	1.59	0.92 $\pm$.04	0.94 $\pm$.04	0.94 $\pm$.04	0.96 $\pm$.04
μ — slot 0	3.02	2.39 $\pm$.12	2.39 $\pm$.13	2.83 $\pm$.21	2.84 $\pm$.12
σ — slot 0	1.72	0.86 $\pm$.10	0.91 $\pm$.08	0.87 $\pm$.13	0.93 $\pm$.11
a — slot 1	1.60	0.83 $\pm$.02	0.84 $\pm$.05	0.81 $\pm$.04	0.84 $\pm$.05
μ — slot 1	11.09	8.98 $\pm$.15	8.99 $\pm$.11	9.55 $\pm$.24	9.55 $\pm$.33
σ — slot 1	2.30	1.29 $\pm$.04	1.31 $\pm$.03	1.15 $\pm$.01	1.21 $\pm$.10

runs: UNet · set-pred · Hungarian · $K=5$ · param-MSE · full norm ladder · hv flips · patch 64×32 (presence lever = column); five seeded replicas per arm, cells mean $\pm$ seed std; held-out test; means conditioned on the slot's GT activity mask. +A active normalization, +B presence balance, +AB both; the none arm = the $K=5$ set-pred-Hungarian heads arm. **bold** = closest to the GT column, marked only where best and worst differ by more than 5%; **green** = the tail amplitudes B re-brightens, **red** = the tail gates it over-fires. Width rows: prediction std over GT-active pixels vs the GT's own (slots 0–1; tail slots in the report). μ , σ in elevation units; a in linear amplitude.